

Project Title:-

Suspicious PowerShell Network Connection Detected

Executive Summary:-

Investigated a PowerShell-initiated outbound network connection using Sysmon Event ID 3 and Splunk. Analysis confirmed an HTTP connection from powershell.exe to an external destination IP address. The activity was generated in a controlled lab environment to demonstrate network connection monitoring and investigation techniques.

Environment:-

- Tool Used: Sysmon
- Data Source: Sysmon Event ID 3 (Network Connection)
- Platform: Splunk
- Host: Windows 10 Lab VM

Alert Overview:-

- A PowerShell process initiated an outbound HTTP connection to an external IP address. The activity was reviewed to determine whether the connection represented legitimate administrative activity or potential malicious behavior.

Investigation Process:-

- Reviewed Sysmon Network connection logs
- Investigated destination path targeted
- Reviewed PowerShell execution flags

Response Actions:-

- Preserved evidence
- Network Connection details documented
- Destination IP reviewed
- Additional PowerShell activity investigated
- Finding recorded for reporting

Key Findings:-

- Destination IP: 104.20.23.154
- Destination Port: 80
- Image: powershell.exe

Indicators of Compromise (IOCs):-

Process: powershell.exe

User: SOC-Lab

Host: DESKTOP-6A40QRO

Destination IP: 104.20.23.154

Destination Port: 80

Source IP: 192.168.137.131

Protocol: HTTP

Timeline:-

18:32:22 – Network connection request created
18:33:12 – Alert reviewed in splunk
18:33:35 – Network activity analysis performed
18:34:22 – PowerShell activity reviewed
18:35:28 – Network request Blocked
18:37:02 – Additional persistence mechanism reviewed
18:40:00 – Investigation closed

Detection Logic:-

```
index=* source="WinEventLog:Microsoft-Windows-Sysmon/Operational" "EventID>3<"  
| rex "Image">(?"Image">[^\<]+)"  
| rex "User">(?"User">[^\<]+)"  
| rex "DestinationIp">(?"DestinationIp">[^\<]+)"  
| rex "DestinationPort">(?"DestinationPort">[^\<]+)"
```

MITRE ATT&CK Mapping:-

- MITRE ATT&CK T1105 = Ingress Tool Transfer
- MITRE ATT&CK T1059.001 = PowerShell

Lessons Learned:-

- Through this case and investigation, I learned that Network Connection logs may not always be available due to configuration or installation of tool file, before shift making sure all the needed file are available as per requirement.

Skills Demonstrated:-

- Log Analysis
- Splunk
- Incident Response
- Threat Hunting
- Documentation

Severity:-

Low

Reason is, PowerShell established an outbound HTTP connection to an external destination. No evidence of malware execution, persistence creation, credential access or command-and-control activity was identified.

Evidence Collected:-

1. Event ID 3 Detection

6/8/26
6:32:22.991 PM

<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event"><System<Provider Name="Microsoft-Windows-Sysmon" Guid="{5770385f-c22a-43e0-bf4c-06f5698ffbd9}" /><EventID>3</EventID><Version>5</Version><Level>4</Level><Task>3</Task><Opcode>8</Opcode><Keywords>8x8000000000000000</Keywords><TimeCreated SystemTime="2026-06-08T17:32:22.9916099Z" /><EventRecordID>51956</EventRecordID><Correlation><Execution ProcessID="3136" ThreadID="5988" /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>DESKTOP-6A40QRO</Computer><Security UserID="S-1-5-18" /></System><EventData><Data Name="RuleName"></Data><Data Name="UtcTime">2026-06-08 17:32:20.766</Data><Data Name="ProcessGuid">{e1bdf9d-e9d-fb66-6a26-7502-000000004600}</Data><Data Name="ProcessId">11984</Data><Data Name="Image">C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name="User">DESKTOP-6A40QRO\OC-Lab</Data><Data Name="Protocol">tcp</Data><Data Name="Initiated">true</Data><Data Name="SourceIsIpv6">false</Data><Data Name="SourceIp">192.168.137.131</Data><Data Name="SourceHostname">DESKTOP-6A40QRO.localdomain</Data><Data Name="SourcePort">50147</Data><Data Name="SourcePortName"></Data><Data Name="DestinationIsIpv6">false</Data><Data Name="DestinationIp">104.20.23.154</Data><Data Name="DestinationHostname"></Data><Data Name="DestinationPort">80</Data><Data Name="DestinationPortName">http</Data></EventData></Event>

Event Actions

Type	Field	Value	Actions
Selected	host	DESKTOP-6A40QRO	
	source	WinEventLog:Microsoft-Windows-Sysmon/Operational	
	sourcetype	XmlWinEventLog:Microsoft-Windows-Sysmon/Operational	
Event	Image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	
	User	DESKTOP-6A40QRO\SOC-Lab	
Time	_time	2026-06-08T18:32:22.991+01:00	
Default	index	main	
	linecount	1	
	punct	<_=":/:./ \\\\/><_=":/:./ \\\\/><_=":/:./ \\\\/><_=":/:./ \\\\/>	
	splunk_server	LAPTOP-CHVD1ONG	

2. Destination IP & Destination Port

6/8/26
6:32:22.991 PM

<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event"><System<Provider Name="Microsoft-Windows-Sysmon" Guid="{5770385f-c22a-43e0-bf4c-06f5698ffbd9}" /><EventID>3</EventID><Version>5</Version><Level>4</Level><Task>3</Task><Opcode>8</Opcode><Keywords>8x8000000000000000</Keywords><TimeCreated SystemTime="2026-06-08T17:32:22.9916099Z" /><EventRecordID>51956</EventRecordID><Correlation><Execution ProcessID="3136" ThreadID="5988" /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>DESKTOP-6A40QRO</Computer><Security UserID="S-1-5-18" /></System><EventData><Data Name="RuleName"></Data><Data Name="UtcTime">2026-06-08 17:32:20.766</Data><Data Name="ProcessGuid">{e1bdf9d-e9d-fb66-6a26-7502-000000004600}</Data><Data Name="ProcessId">11984</Data><Data Name="Image">C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name="User">DESKTOP-6A40QRO\SOC-Lab</Data><Data Name="Protocol">tcp</Data><Data Name="Initiated">true</Data><Data Name="SourceIsIpv6">false</Data><Data Name="SourceIp">192.168.137.131</Data><Data Name="SourceHostname">DESKTOP-6A40QRO.localdomain</Data><Data Name="SourcePort">50147</Data><Data Name="SourcePortName"></Data><Data Name="DestinationIsIpv6">false</Data><Data Name="DestinationIp">104.20.23.154</Data><Data Name="DestinationHostname"></Data><Data Name="DestinationPort">80</Data><Data Name="DestinationPortName">http</Data></EventData></Event>

3. Splunk Query

```
1 index=* source="WinEventLog:Microsoft-Windows-Sysmon/Operational" DestinationIp
2 | rex "CommandLine">\s*(?<CommandLine>[^\s]+)"
3 | rex "ParentImage">\s*(?<ParentImage>[^\s]+)"
4 | rex "Image">\s*(?<Image>[^\s]+)"
5 | rex "User">\s*(?<User>[^\s]+)"
```

4. PowerShell execution

Type	Field	Value	Actions
Selected	host	DESKTOP-6A40QRO	
	source	WinEventLog:Microsoft-Windows-Sysmon/Operational	
	sourcetype	XmlWinEventLog:Microsoft-Windows-Sysmon/Operational	
Event	Image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	
	User	DESKTOP-6A40QRO\SOC-Lab	
Time	_time	2026-06-08T18:32:22.991+01:00	
Default	index	main	
	linecount	1	
	punct	<_=":/:./ \\\\/><_=":/:./ \\\\/><_=":/:./ \\\\/><_=":/:./ \\\\/>	
	splunk_server	LAPTOP-CHVD1ONG	

Analyst Verdict:-

PowerShell established an outbound HTTP connection to an external destination IP address. Investigation confirmed the activity was generated during a controlled lab exercise using Invoke-WebRequest. No malicious payload delivery, persistence mechanism or command-and-control activity was identified. Activity was classified as benign laboratory-generated traffic.